Stability and comparability of meritocratic beliefs in school-age students: A measurement invariance approach across time and cohorts

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Abstract

Meritocratic beliefs shape how adolescents understand inequality and the legitimacy of social arrangements. Schools are key sites of meritocratic socialization, making adolescence a crucial period for studying how these beliefs are formed. However, research on adolescents' meritocratic beliefs has often relied on unidimensional measures that conflate perceptions of how allocation works with preferences about how it should work and treat meritocratic and non-meritocratic principles as opposites. This study tests a multidimensional model that distinguishes between perceived and preferred meritocracy and non-meritocracy. Using two-wave panel survey data from Chilean students in 8th and 10th grade (Wave 1: N = 846; Wave 2: N = 662), we assess whether these dimensions can be distinguished, compared across school stages, and measured equivalently over time. Results support a four-factor structure and show that endorsement of merit can coexist with recognition of non-meritocratic advantage. The scale achieves strict longitudinal invariance within students but not across school levels.
Keywords: Meritocracy · Schools · Measurement invariance · Socialization · Chile

1 Introduction

Despite rising economic inequality and limited social mobility in contemporary societies (Chancel et al., 2025; López-Roldán & Fachelli, 2021), the belief that inequalities reflect disparities in effort and talent instead of opportunities remains remarkably widespread among citizens (Mijs, 2021). This idea refers to meritocracy, the notion that social rewards are, and should be, allocated according to individual effort and ability rather than social origins or personal connections (Young, 1958). Although the term "meritocracy" was originally coined as a critique of a future society where meritocratic principles would be taken to an extreme (Young, 1958), it has since been widely adopted as a normative ideal that justifies existing social arrangements and motivates individual striving (Mijs, 2021; Sandel, 2020). On the one hand, meritocracy can be seen as a powerful narrative that promotes social cohesion and motivates individuals to work hard and pursue their goals. On the other hand, it can also be criticized for obscuring structural inequalities and legitimizing the status quo (Mijs, 2021; Sandel, 2020). Therefore, the persistence of meritocratic beliefs could have important implications for how people understand and respond to inequality, as well as for the legitimacy of social institutions (Madeira et al., 2019; Sandel, 2020).

The idea of meritocracy is particularly salient in school settings, where it operates not only as a cultural ideal but also as a central organizing principle of institutional life. Schools promote meritocracy as a normative standard by presenting education as a legitimate pathway to upward mobility and by framing individual effort and achievement as the proper basis of social rewards (Lampert, 2013; Van De Werfhorst, 2024). In this environment, students encounter meritocracy not only as discourse but also as a lived experience through grading, selection, and tracking, which make merit visible and consequential (Darnon et al., 2018; Resh & Sabbagh, 2014; Tang et al., 2025; Traini, 2022). In this same socialization process, students are also exposed to non-meritocratic factors, such as the influence of family background, social connections, and structural barriers, which may coexist with or undermine the meritocratic ideal (Bourdieu & Passeron, 1990; Goudeau & Croizet, 2017; Zhou, 2019). In this

line, extant research shows that a stronger endorsement of meritocratic beliefs in school is associated with greater justification of inequality (Batruch et al., 2022; Lampert, 2013; Wiederkehr et al., 2015), even when students may simultaneously recognize structural constraints, reflecting a form of dual consciousness (C. Liu & Wang, 2025). Taken together, these tensions make schools a crucial setting for examining how adolescents understand meritocracy, non-meritocratic constraints, and the fairness of unequal outcomes.

The present paper contributes to this line of research by examining how school-aged students perceive and endorse meritocracy. We identify two main contributions of this study. The first one is the conceptualization and measurement of meritocratic beliefs in school settings. Most of the research in this area has focused on the consequences of meritocratic beliefs, while less attention has been paid to their conceptualization and measurement. This is a crucial gap in the extant literature, as how meritocratic beliefs are conceptualized and measured can have important implications for the validity of research findings and their interpretation. Addressing this point, we approach the study of meritocratic beliefs at school from a multidimensional approach (Castillo et al., 2023), which distinguishes between perceptions (descriptive beliefs about how meritocracy works) and preferences (normative beliefs about how it should work), as well as between meritocratic principles (e.g., effort and ability) and non-meritocratic principles (e.g., family background and social connections). By adopting this perspective, we can better capture the complexity of adolescents' beliefs about meritocracy and the role schools play in fostering them.

A second contribution of this study is to examine the stability and comparability of meritocratic beliefs across school stages and over time. Most research on meritocratic beliefs remains cross-sectional, limiting what we can infer about how these orientations develop through school-based socialization during adolescence. This matters because adolescence is precisely when reasoning about justice and inequality becomes more sophisticated and when school-based sorting mechanisms become increasingly salient (Henry & Saul, 2006; Resh & Sabbagh, 2014), potentially reshaping how students interpret meritocratic narratives in light of lived constraints (Allen, 2016). To address this, we test measurement invariance across two cohorts of Chilean secondary students (8th and 10th graders) and longitudinally within the same students across two waves (2023–2024). Establishing invariance is essential for valid comparisons: without it, observed differences across cohorts or over time may reflect changes in how items are understood and used rather than substantive change in the underlying constructs (Putnick & Bornstein, 2016; Van De Schoot et al., 2012; Van De Schoot et al., 2015). By providing this measurement foundation, the study clarifies whether apparent differences in meritocratic beliefs reflect genuine developmental change or measurement non-equivalence, and it strengthens subsequent analyses of how these beliefs relate to different attitudes toward inequality and social mobility.

According to these two contributions, the research questions guiding this study are as follows:

(1) Do school-aged students distinguish between meritocratic and non-meritocratic principles, as well as between perceptions ("what is") and preferences ("what should be")?

(2) Can these dimensions be measured in ways that are comparable across age cohorts and stable over time?

We argue that a multidimensional conception of meritocratic beliefs clarifies what adolescents actually endorse—descriptions, ideals, or both—and provides a stronger basis for valid developmental and longitudinal analysis. In the following sections, we first review the literature on meritocratic beliefs in school settings, highlighting the conceptual and methodological gaps that

this study aims to address. Then, we present the methodology of our study, including the sample, measures, and analytical strategy. Finally, we discuss the implications of our findings for understanding adolescents' beliefs about meritocracy and the consequences of these beliefs for attitudes toward inequality.

2 Theoretical and empirical background

2.1 What is meritocracy, and how is it measured?

Meritocracy refers to a distributive system in which individual merit—typically effort and ability—is treated as the main criterion for allocating rewards, rather than social origins or inherited privilege (Bell, 1972; Young, 1958). Although Young (1958) introduced the term as a dystopian critique, it has since been reappropriated as a positive ideal of fairness, especially in liberal and market-oriented societies (Mijs, 2021; Van De Werfhorst, 2024). Sociologically, meritocracy operates both as a cognitive judgment about how inequality works and as a moral lens through which inequality is evaluated (Castillo et al., 2019; Heuer et al., 2020). Precisely because it frames outcomes as earned, however, it can also reinforce inequality by encouraging winners to see their position as deserved and losers to internalize failure rather than question structural conditions (García-Sierra, 2023; Mijs, 2016a; Sandel, 2020).

Research on adults has examined both the determinants and consequences of meritocratic perceptions. Higher-status and upwardly mobile individuals are more likely to endorse merit-based explanations of social outcomes (Duru-Bellat & Tenret, 2012; García-Sánchez et al., 2018; Mijs et al., 2022; Traini et al., 2025), whereas perceiving society as meritocratic is associated with lower support for redistribution, greater acceptance of inequality, and stronger endorsement of market-based allocation (Castillo et al., 2025, 2019; Hoyt et al., 2023; Pañeda-Fernández et al., 2026; Tejero-Peregrina et al., 2025). Experimental evidence is consistent with a causal pathway in which upward-mobility or self-made narratives strengthen meritocratic perceptions and reduce redistributive support, while downward-mobility cues produce the opposite pattern (Deng & Wang, 2025; Matamoros-Lima et al., 2025). Overall, this literature suggests that meritocratic perceptions are systematically linked to both social position and orientations toward inequality.

This research agenda faces important conceptual and measurement challenges. Some studies define meritocracy narrowly, often reducing it to effort or simplified readings of Young's formulation (Mijs, 2021; Wiederkehr et al., 2015; Young, 1958), while others blur it with broader constructs such as system justification, beliefs about mobility, or support for equal opportunity (Batruch et al., 2022; Darnon et al., 2018; Day & Fiske, 2017; Deng & Wang, 2025; Jost et al., 2004; McCoy & Major, 2007). As a result, conceptually distinct orientations are often collapsed into a single indicator, and the distinction between descriptive perceptions of how rewards are allocated and normative preferences about how they should be allocated is acknowledged but rarely implemented analytically.

A related limitation concerns the treatment of non-meritocratic factors such as family wealth, social connections, or inherited advantage. As Castillo et al. (2023) notes, many studies ask respondents to rate the importance of meritocratic and non-meritocratic factors for "getting ahead," but then collapse them into a single score. This is especially problematic when researchers treat both as opposite ends of the same continuum, for example, through subtraction scores (Reynolds & Xian, 2014). Such measures impose a zero-sum assumption—namely, that endorsing merit implies rejecting structural or relational

advantages—and therefore rule out the possibility that both beliefs coexist. Recent debates and empirical evidence suggest the opposite: support for merit can coexist with recognition of privilege, contacts, or structural constraints (Kwon & Pandian, 2024; C. Liu & Wang, 2025; Mijs, 2026; Tang et al., 2025; Wiesner & Sachweh, 2026; Zhu, 2025). The implication is clear: unidimensional indices, especially subtraction scores, can obscure substantively distinct belief profiles.

Responding to these limitations, Castillo et al. (2023) proposed a multidimensional framework that distinguishes meritocratic beliefs along two analytically independent axes: perceptions versus preferences, and meritocratic versus non-meritocratic allocation principles. Preferences capture normative ideals about how rewards should be distributed, whereas perceptions capture descriptive evaluations of how society actually works (Janmaat, 2013). At the same time, meritocratic elements (e.g., effort, talent) and non-meritocratic elements (e.g., social origins, inherited privilege, networks) are treated as distinct components rather than opposite poles of a single continuum. This yields four non-redundant constructs—meritocratic preferences, meritocratic perceptions, non-meritocratic preferences, and non-meritocratic perceptions—and allows combinations that older measures collapse, such as endorsing meritocracy as an ideal while recognizing the role of family background and connections. The framework is also parsimonious and therefore well suited to large-scale surveys, although its reliance on only two items per factor may constrain reliability and content coverage.

Evidence from adult samples using confirmatory factor analysis suggests that respondents can differentiate among these four dimensions and that they are systematically related. In particular, stronger perceptions that non-meritocratic factors are rewarded tend to coexist with stronger preferences for meritocratic allocation. This pattern aligns with Zhu's (2025) notion of "dual consciousness," understood here as the coexistence of meritocratic and non-meritocratic beliefs, as well as descriptive assessments ("is") and legitimacy judgments ("should").

In school research, these measurement challenges have received little attention. One recent exception is Chauvin et al. (2026), who proposes a multidimensional measure of school meritocracy, but it applies only to teachers and remains limited to perceived meritocracy. Another relevant study, by Castillo et al. (2024), applies the multidimensional framework to Chilean 10th graders and shows that students can differentiate between perceptions and preferences, as well as between meritocratic and non-meritocratic principles. However, that study does not examine whether these dimensions are stable across cohorts or over time, limiting our understanding of their developmental trajectory and comparability.

2.2 The socialization of meritocracy at school

Socialization refers to the process through which individuals internalize norms, values, and practices, acquiring the dispositions needed to function in society. From a sociological perspective, this process is neither neutral nor purely individual, but structured by relations of power and inequality (Bourdieu, 2007). Likewise, social learning approaches emphasize that children adopt behavioral patterns and value systems modeled in their everyday environments, especially within families (Bandura, 1969). The internalization of meritocratic beliefs—such as the idea that effort and talent determine success—should therefore be understood not as a purely rational judgment but as the outcome of structured, socially differentiated socialization processes.

Families play a foundational role in early meritocratic orientations. Parenting often reflects class-based strategies of status reproduction (Bernstein, 2005; Bourdieu & Passeron, 1990), and children internalize parental views of success, fairness, and

responsibility through everyday interactions (Bandura, 1969). Yet schools constitute a particularly consequential institutional arena in which these beliefs are further shaped, reinforced, and contested.

Schools socialize meritocracy both culturally and institutionally. They promote narratives of effort, achievement, self-improvement, and education as a legitimate route to mobility (Batruch et al., 2022; Chauvin et al., 2026; Darnon et al., 2018; Mijs, 2021; Wiederkehr et al., 2015), while embedding these ideals in routine practices of evaluation and classification such as grading, tracking, selection, awards, and credentials (Resh & Sabbagh, 2014; Traini, 2022). At the same time, schools are settings where non-meritocratic advantages—such as family resources, cultural capital, peer environments, and unequal school quality—also become visible and consequential (Batruch et al., 2022; Resh & Sabbagh, 2014). This duality makes educational settings especially important for understanding how meritocratic beliefs are formed.

A growing literature shows that this duality can generate internally differentiated orientations. Students may endorse meritocracy as a normative ideal while simultaneously recognizing structural constraints, a pattern described as "dual consciousness" (C. Liu & Wang, 2025; Tang et al., 2025). More broadly, meritocratic framing encourages students to interpret success and failure in individual terms, reinforcing personal responsibility while limiting attention to structural conditions (Darnon et al., 2018; Lampert, 2013). Even disadvantaged students often adopt meritocratic narratives, suggesting that prolonged exposure to school-based evaluative environments is a powerful mechanism shaping beliefs about justice and inequality (Henry & Saul, 2006).

These beliefs matter beyond school. Perceiving schools as meritocratic—particularly believing that effort and talent are fairly rewarded—has been linked to greater legitimation of inequality, including stronger system justification, lower support for redistribution, and greater acceptance of unequal access to key social services (Batruch et al., 2022; Castillo et al., 2024; Wiederkehr et al., 2015). Schools thus shape not only academic trajectories, but also early distributive beliefs that may normalize broader social inequalities.

2.3 This study

This study examines how students understand meritocracy during a formative stage of socialization—schooling—and whether these understandings can be measured reliably and comparably. Building on Castillo et al.'s (2023) conceptual and measurement framework, we test whether students differentiate between perceptions of how meritocracy operates ("what is") and preferences about how it should operate ("what ought to be"), and whether these judgments distinguish meritocratic elements (effort, talent) from non-meritocratic ones (family wealth, personal contacts). Beyond factorial validity, the paper focuses on measurement stability—whether the same latent constructs are captured comparably across school stages and over time within the same students.

Establishing measurement invariance is essential because it provides the empirical foundation for valid comparisons (Chen, 2007; Putnick & Bornstein, 2016). Without it, observed differences between groups or across time may reflect changes in how the instrument functions rather than substantive differences in the underlying constructs (Putnick & Bornstein, 2016; Van De Schoot et al., 2015). In sociological terms, invariance matters because comparisons are only interpretable if a given latent score carries the same meaning across ages or time points; otherwise, apparent developmental or group differences may conflate substantive variation with measurement artifacts (Y. Liu et al., 2017; Van De Schoot et al., 2012).

The study uses longitudinal data from two cohorts of Chilean secondary students: 8th graders (approximately 12–13 years old) and 10th graders (approximately 15–16 years old) at baseline, followed across two waves. These cohorts are theoretically strategic because they capture distinct stages of adolescence, when reasoning about justice and inequality becomes more sophisticated and school-based sorting mechanisms—grades, tracking, and peer comparisons—become increasingly salient (Henry & Saul, 2006; Resh & Sabbagh, 2014). Developmental research further suggests that by around age 10, children can already articulate judgments about social differences, often emphasizing individual, merit-based explanations while beginning to incorporate opportunity-based accounts (Imhoff & Brussino, 2025; Sigelman, 2013). As students move through the school system, cumulative exposure to evaluation and selection may make the limits of effort as a sufficient explanation for success more visible, generating a potential "reality shock" as meritocratic discourse clashes with experienced inequality and constraint (Allen, 2016). Both cohorts are embedded in the same highly stratified educational system, yet differ in their cumulative exposure to schooling and to the meritocratic narratives the system promotes. We therefore test invariance across cohorts and waves to determine whether observed differences and changes can be interpreted substantively rather than as artifacts of measurement non-equivalence.

2.3.1 Hypotheses

H_1 **(Dimensional differentiation)**. Students' meritocratic beliefs are best represented by four distinct factors: perceived meritocracy, perceived non-meritocracy, preferred meritocracy, and preferred non-meritocracy.

H_2 **(Between-cohort measurement invariance)**. The four-factor model of meritocratic beliefs is invariant across age cohorts.

H_3 **(Longitudinal measurement invariance)**. The four-factor model of meritocratic beliefs is invariant across waves within students.

3 Method

3.1 Participants

This study uses secondary data from the 2023 and 2024 waves of the Panel Survey on Education and Meritocracy (EDUMER), which examines school-age students' beliefs, attitudes, and behaviors regarding meritocracy, inequality, and citizenship[1]. The questionnaire was administered to sixth-grade and first-year secondary students in nine schools in Chile's Metropolitan and Valparaíso regions. Although the sample was non-probabilistic and lacked quota controls, a minimum target of 900 students was set to ensure adequate statistical power. After data processing and listwise deletion, the analytical sample included 846 students in Wave 1 (386 girls, 421 boys, 39 identifying as other; M_{age} = 13.4, SD_{age} = 1.6) and 662 in Wave 2 (303 girls, 338 boys, 21 identifying as other; M_{age} = 14.4, SD_{age} = 1.6), yielding a retention rate of 78.3% (attrition = 21.7%). The main sources of attrition were school transfers between academic years and high absenteeism in several schools.

[1]The survey was collected within the research project "Meritocracy in schools: Moral foundations of the education market and its implications for civic education in Chile" (FONDECYT Regular 2021 Nº 1210847).

3.2 Procedure

Data were collected by a professional research firm using online CAWI questionnaires. All participants received informed consent, which was reviewed and approved by a parent or legal guardian. To enable longitudinal linkage while protecting confidentiality, the survey firm generated anonymous unique identifiers for each student, and no directly identifying information was included in the analytic files.

3.3 Measures

3.3.1 Scale of perceptions and preferences of meritocracy

Meritocratic and non-meritocratic perceptions and preferences were measured using the same items proposed in the original scale (Castillo et al., 2023). Each of the four dimensions of the scale is measured by two items. The dimensions are:

- Perceived meritocracy: the extent to which effort and ability are rewarded in Chile,
- Perceived non-meritocracy: the extent to which success is perceived as linked to connections and family wealth,
- Preference for meritocracy: agreement with that those who work harder or are more talented should be better rewarded,
- Preference for non-meritocracy: agreement with that it is acceptable for individuals with better connections or wealthy parents to achieve greater success (see Table 1).

Each item is rated on a four-point Likert scale ranging from "strongly disagree" (1) to "strongly agree" (4), therefore higher scores indicate stronger endorsement of the corresponding perception or preference.

3.3.2 Cohort level

To distinguish students' academic level across waves for conditional and multi-group invariance analyses, we created a cohort indicator capturing whether the respondent belonged to the primary (sixth grade) or secondary (first-year secondary) cohort at the time of each survey wave. Descriptive statistics[2] by cohort are reported in Table 2.

3.4 Analytical strategy

3.4.1 Measurement model and estimation

To evaluate the underlying structure of the scale, we employed Confirmatory Factor Analysis (CFA) based on a measurement model with four latent factors, using Diagonally Weighted Least Squares with robust correction (WLSMV) estimation. This estimator is particularly suitable for ordinal data, such as four-point Likert-type scales, as it avoids the bias associated with treating categorical data as continuous (Kline, 2023).

[2] Table 2 totals (N = 1,508) correspond to the sum of person-wave observations (846 at T1 + 662 at T2), rather than unique individuals.

Table 1: Items for perceptions and preferences for meritocracy scale

Components	Dimensions	Item (English)	Item original (Spanish)
Perception	Meritocratic	In Chile people are rewarded for their efforts	En Chile las personas son recompensadas por sus esfuerzos
		In Chile people are rewarded for their intelligence and ability	En Chile las personas son recompensadas por su inteligencia y habilidad
	Non meritocratic	In Chile those with wealthy parents do much better in life	En Chile a quienes tienen padres ricos les va mucho mejor en la vida
		In Chile those with good contacts do much better in life	En Chile a quienes tienen buenos contactos les va mejor en la vida
Preference	Meritocratic	Those who work harder should reap greater rewards than those who work less hard	Quienes más se esfuerzan deberían obtener mayores recompensas que quienes se esfuerzan menos
		Those with more talent should reap greater rewards than those with less talent	Quienes poseen más talento deberían obtener mayores recompensas que quienes poseen menos talento
	Non meritocratic	It is good that those who have rich parents do better in life	Está bien que quienes tengan padres ricos les vaya mejor en la vida
		It is good that those who have good contacts do better in life	Está bien que quienes tengan buenos contactos les vaya mejor en la vida

Table 2: Descriptive statistics for cohort level

			Age		Gender		
Wave	Cohort level	N	Mean	SD	Man	Women	Others
Wave 1	Primary	406	11.85	0.56	195	191	20
	Secondary	440	14.79	0.80	226	195	19
	Total	846	13.38	1.63	421	386	39
Wave 2	Primary	319	12.83	0.55	161	145	13
	Secondary	343	15.76	0.75	177	158	8
	Total	662	14.35	1.61	338	303	21

Model fit was assessed following the guidelines of Brown (2015), using several indices: the Comparative Fit Index (CFI) and the Tucker-Lewis Index (TLI), with acceptable values above 0.95; the Root Mean Square Error of Approximation (RMSEA), with values below 0.06 indicating good fit; and the Chi-square statistic (acceptable fit indicated by $p > 0.05$ and a Chi-square/df ratio < 3).

3.4.2 Measurement invariance

A key contribution of this study is its assessment of measurement stability through factorial invariance testing (Davidov et al., 2014). We implemented two complementary strategies: cross-group invariance across school levels (primary vs. secondary students) and longitudinal invariance across two panel waves. Given the ordinal nature of the items, we used estimation procedures appropriate for ordered categorical indicators, following Y. Liu et al. (2017) for longitudinal analyses and Svetina et al. (2020) for cross-group tests. We assessed invariance using a hierarchical sequence of nested confirmatory factor analysis models that imposed progressively stricter equality constraints. For longitudinal invariance, we estimated a configural model specifying the same factor structure over time while allowing item parameters to vary, followed by metric, scalar, and strict models constraining factor loadings, thresholds/intercepts, and residual variances, respectively. For cross-group invariance, we estimated a configural model, then a threshold-invariant model, and finally a model jointly constraining thresholds and factor loadings across groups.

In addition to the chi-square difference test, we assessed invariance using changes in approximate fit indices. Following Chen (2007), we treated ΔCFI as the primary criterion and ΔRMSEA as supplementary, indicating noninvariance when $\Delta\text{CFI} \leq -0.010$ and $\Delta\text{RMSEA} \geq 0.015$.

All analyses were performed using the *lavaan* package in R version 4.5.2. The hypotheses of this research were pre-registered on the Open Science Framework (OSF).

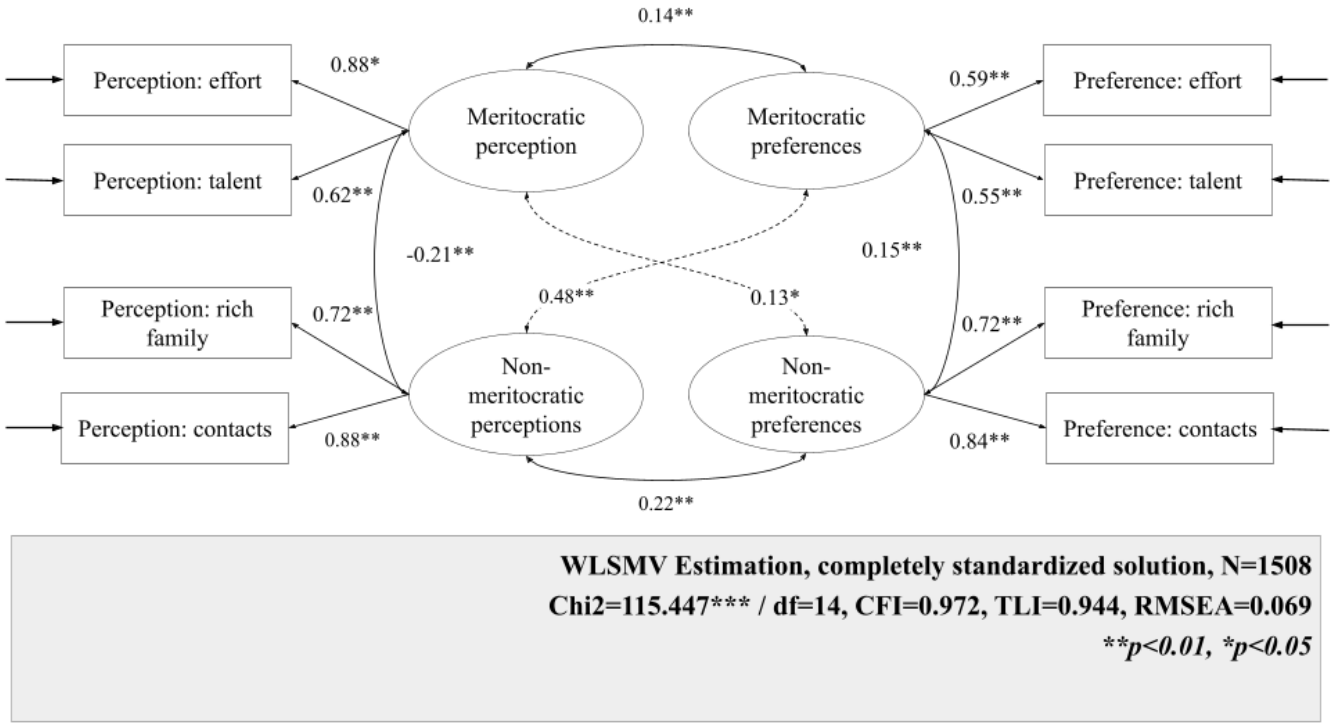
4 Results

4.1 Measurement model

Figure 1 presents the standardized factor loadings and the overall fit indices of the general measurement model estimated on the pooled analytical sample across cohorts and waves. The model shows a reasonable, though not fully satisfactory, fit to the data (CFI = 0.97; TLI = 0.94; RMSEA = 0.07). Although the CFI and RMSEA fall within commonly used acceptable ranges, the TLI remains below conventional thresholds, indicating that fit is acceptable in some respects but not uniformly strong across indices. Overall, these results suggest that the proposed four-factor structure captures the main covariance pattern in the combined sample reasonably well.

All standardized factor loadings are statistically significant and generally moderate to strong, though their magnitude varies across dimensions. For meritocratic perceptions, both indicators load positively on the latent factor, with effort showing a particularly strong loading (β = 0.88) and talent a moderately strong loading (β = 0.62). Meritocratic preferences exhibit more even, but somewhat weaker, loadings for effort (β = 0.59) and talent (β = 0.55). For non-meritocratic perceptions, both

Figure 1: Multigroup model diagram



indicators load strongly, especially contacts (β = 0.88), followed by coming from a rich family (β = 0.72). Non-meritocratic preferences also show strong and relatively balanced loadings, with contacts (β = 0.84) and rich family (β = 0.72), indicating that both indicators are similarly representative of this latent dimension.

The latent correlations reveal a differentiated structure among the four dimensions. Meritocratic perceptions and meritocratic preferences are positively correlated (r = 0.14), consistent with some alignment between what students perceive is rewarded and what they believe should be rewarded. Meritocratic and non-meritocratic perceptions are negatively correlated ($r = -0.21$), suggesting that students who perceive greater meritocracy in society tend to view a weaker role for connections and family wealth. At the same time, non-meritocratic perceptions are positively related to meritocratic preferences (r = 0.48, strongest correlation), suggesting that perceiving society as rewarding contacts and wealth may coincide with stronger endorsement of meritocratic ideals. Finally, meritocratic and non-meritocratic preferences are modestly positively correlated (r = 0.15), and non-meritocratic perceptions and preferences are also positively associated (r = 0.22), indicating that recognizing non-meritocratic mechanisms in practice is linked to a greater acceptance of their legitimacy, albeit to a moderate extent.

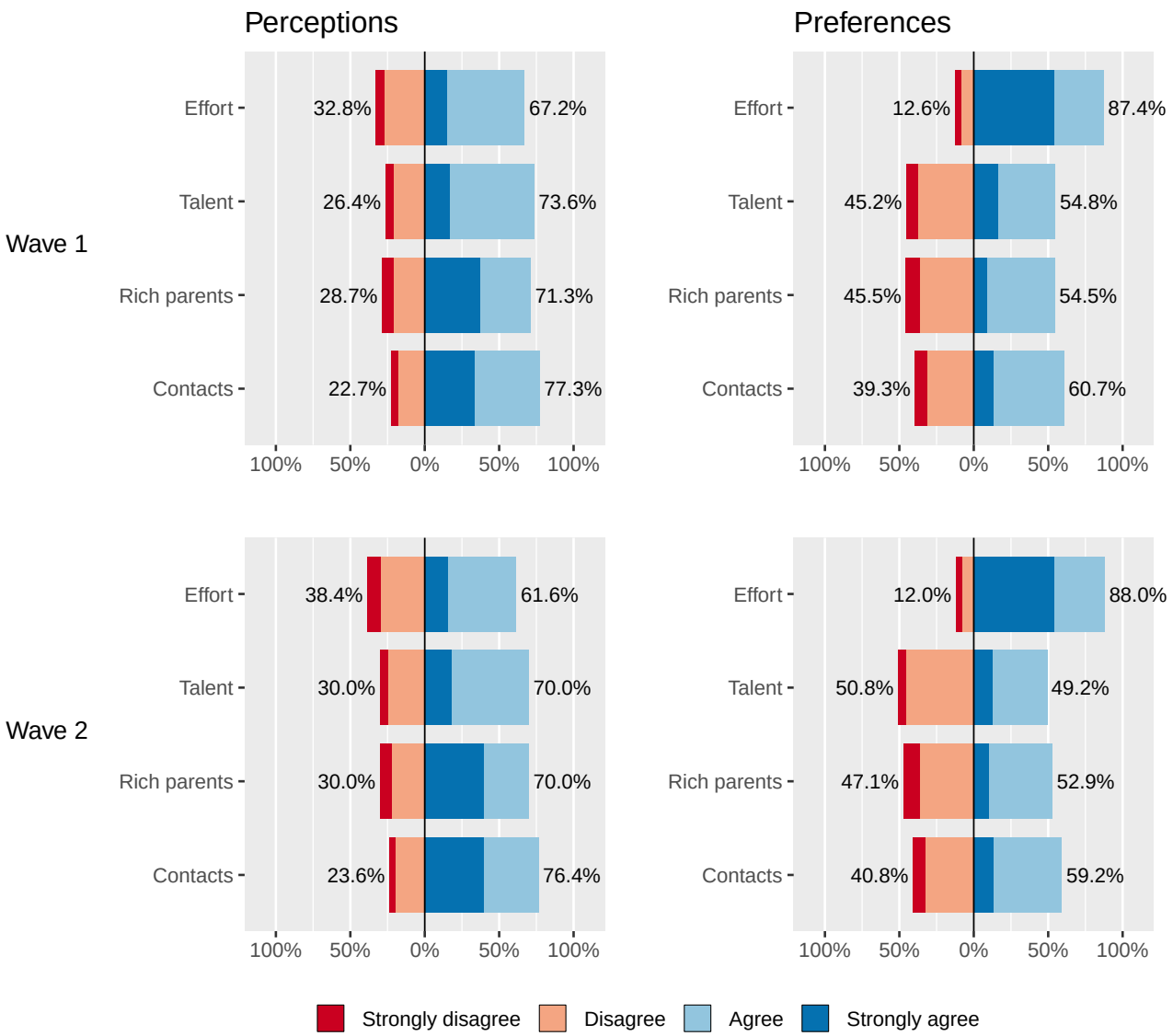
4.2 Invariance analysis

4.2.1 Longitudinal invariance

Figure 2 displays the response distributions for the meritocracy scale, distinguishing perceptions (Panel A) from preferences (Panel B) across waves. Across both waves, a clear and stable gap emerges between how students perceive social rewards to operate and how they think they should operate. In the perception items, agreement is consistently high for contacts,

parental wealth, and talent, whereas effort is also widely endorsed but to a lesser extent, indicating that students recognize both meritocratic and non-meritocratic determinants of success and view talent as more influential than effort. In the preference items, by contrast, effort is overwhelmingly endorsed as the legitimate basis for rewards, far exceeding its perceived role in practice. Talent occupies a more ambivalent normative position, parental wealth remains evenly divided, and contacts are more often accepted than rejected, suggesting that non-meritocratic criteria are not uniformly seen as illegitimate. Wave 2 closely replicates these patterns, with only modest shifts. Overall, the descriptive results reveal a persistent paradox: strong endorsement of effort-centered meritocratic ideals alongside skepticism about their realization in practice.

Figure 2: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by wave



Source: own elaboration based on EDUMER Panel Survey (N Wave 1 = 846; N Wave 2 = 662)

Table 3 reports the results of the longitudinal invariance tests. Overall, the scale achieved the strongest level of invariance (strict invariance) across the two waves. The strict model showed good fit, $\chi^2(84) = 130.17$, CFI = 0.991, RMSEA = 0.031 (90% CI: 0.020–0.040). Moreover, compared with the strong (scalar) model, the deterioration in fit was negligible: $\Delta\chi^2(4)$

Table 3: Longitudinal invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	117.7 (68)	0.991	0.035 (0.024-0.046)	0 (0)	0	0.000	Reference
Weak	122.51 (72)	0.990	0.034 (0.024-0.045)	4.809 (4)	0	-0.001	Accept
Strong	128.53 (80)	0.991	0.032 (0.021-0.042)	6.02 (8)	0	-0.002	Accept
Strict	130.17 (84)	0.991	0.031 (0.02-0.04)	1.635 (4)	0	-0.002	Accept

Table 4: Cohort (multigroup) invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	24.95 (26)	0.993	0.036 (0.009-0.057)	0 (0)	0.000	0.000	Reference
Tresholds	57.56 (42)	0.980	0.049 (0.033-0.064)	43.875 (16) ***	-0.013	0.013	Reject
Tresholds + Loadings	72.35 (46)	0.973	0.053 (0.039-0.067)	17.075 (4) **	-0.006	0.005	Reject

= 1.635, ΔCFI = 0.000, and ΔRMSEA = -0.002. These differences fall well within commonly used cutoffs for invariance evaluation (Chen, 2007), indicating that constraining residual variances in addition to loadings and thresholds/intercepts is tenable. Substantively, this implies that the item–factor relations are stable over time (metric invariance) and that, for a given level of the latent trait, respondents are expected to endorse the same response categories across waves (scalar invariance), supporting the comparability of scores and latent parameters within students over time.

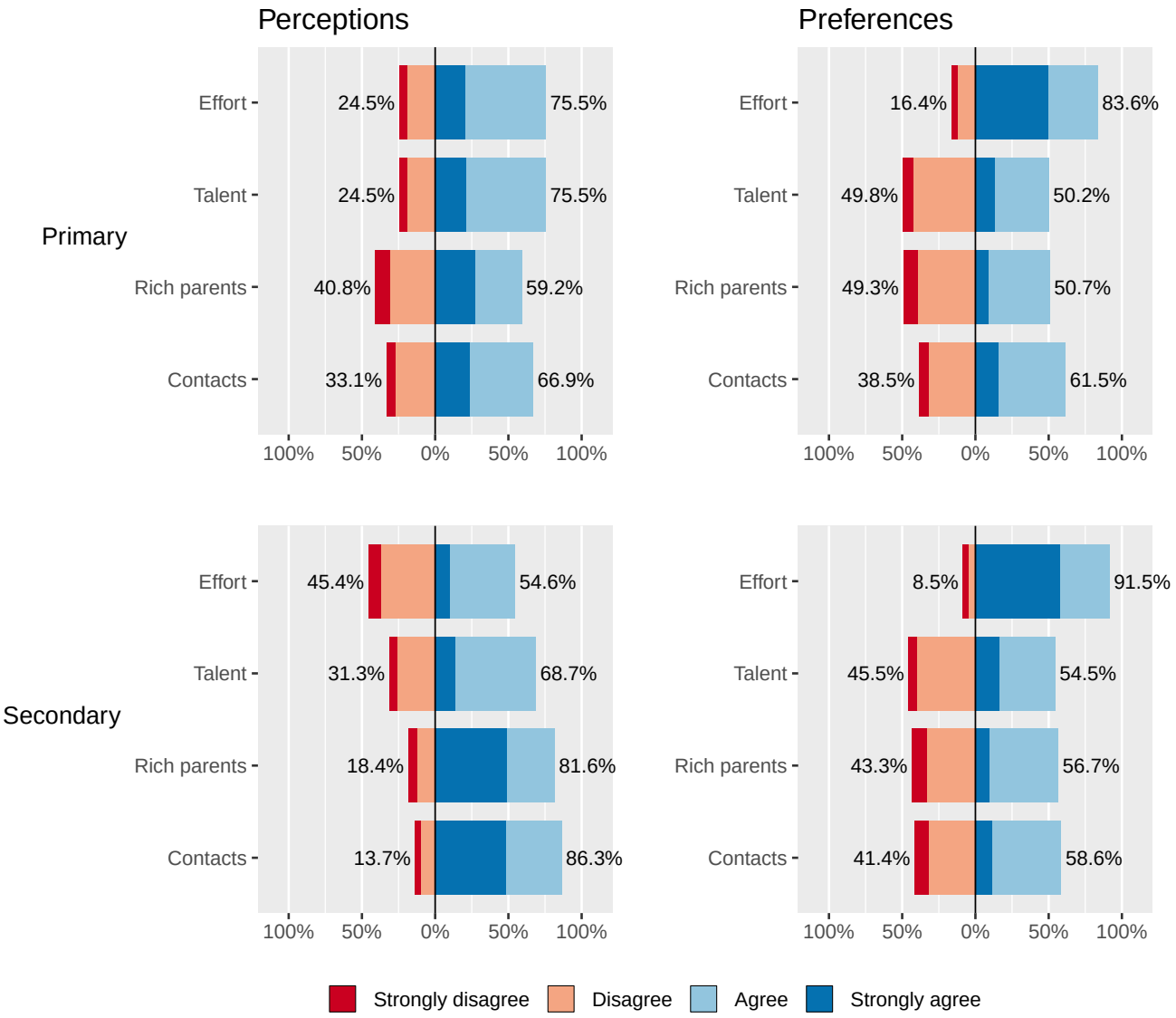
As an additional diagnostic, we used univariate score tests to identify potential sources of localized misfit at each step of the invariance hierarchy. The tests revealed no meaningful violations of the imposed equality constraints, suggesting that model fit was not driven by a small subset of parameters. Global score tests and the smallest p-values at each step are reported in the supplementary materials.

4.2.2 Cohort invariance

Figure 3 shows response distributions by cohort, separating perceptions and preferences. In primary education, effort and talent are perceived as similarly rewarded, while contacts and family wealth are also seen as advantageous, though with more disagreement. In secondary education, perceptions become less meritocratic: agreement that effort is rewarded declines, whereas contacts and family wealth are seen as more strongly shaping success. Preferences are more stable across cohorts. In both groups, effort is the most strongly endorsed legitimate basis for reward, while talent, family wealth, and especially contacts remain more ambivalent. Notably, contacts are more often accepted than rejected in both cohorts. Overall, the cohort comparison reveals a widening gap between an effort-centered meritocratic ideal and the perceived importance of non-meritocratic advantages in practice.

Table 4 summarizes the cross-cohort invariance tests. The scale achieves configural invariance between primary and secondary

Figure 3: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by cohort



Source: own elaboration based on EDUMER Panel Survey (N Primary = 725; N Secondary = 783)

students, but not threshold invariance, and therefore does not meet the prerequisites for stronger forms of invariance (Svetina et al., 2020; Wu & Estabrook, 2016). Constraining thresholds worsened model fit (ΔCFI = −.013; ΔRMSEA = .013), exceeding recommended cutoffs (Chen, 2007). This indicates that response categories do not map equivalently onto the latent constructs across cohorts, limiting direct comparisons of latent means or other latent differences.

To locate the source of cross-cohort non-invariance, we inspected localized misfit in the threshold-invariant model. The largest violations concerned the perceived-effort item, especially the second threshold ($\chi^2 = 10.12, p = .001$) and, to a lesser extent, the first ($\chi^2 = 7.55, p = .006$), suggesting that older students may apply a more critical standard when judging whether effort is rewarded in society.

5 Discussion

The first hypothesis proposed that students' meritocratic beliefs would be best represented by four distinct factors: perceived meritocracy, perceived non-meritocracy, preferred meritocracy, and preferred non-meritocracy. The confirmatory factor analyses support this expectation. Consistent with evidence from adult populations (Castillo et al., 2023), students already distinguish between perceptions and preferences, and between meritocratic and non-meritocratic principles at an early stage of school socialization.

These findings contribute to debates on how meritocratic beliefs should be conceptualized and measured (Kwon & Pandian, 2024; Mijs, 2026; Zhu, 2025). Perceptions of "what is" and preferences about "what should be" are empirically separable among school-aged students, and meritocratic and non-meritocratic principles coexist rather than forming opposite poles of a single continuum. This directly challenges zero-sum measurement strategies, such as subtraction scores, that assume endorsing merit implies rejecting privilege (Reynolds & Xian, 2014; Wiesner & Sachweh, 2026). Instead, our results are more consistent with recent accounts arguing that recognition of non-meritocratic constraints may increase without displacing merit ideals (C. Liu & Wang, 2025; Tang et al., 2025). The positive association between perceived non-meritocracy and stronger meritocratic preferences supports this interpretation, suggesting a normative-reactive response to perceived unfairness (cf. Castillo et al., 2023).

The findings also align with research on meritocracy in educational settings. Schools circulate meritocratic narratives of effort, achievement, and mobility while institutionalizing merit through grading, selection, and tracking (Mijs, 2016b; Resh & Sabbagh, 2014; Tang et al., 2025; Traini et al., 2025). Our results extend this literature by showing that students not only perceive meritocracy in multidimensional terms but also differentiate their preferences and incorporate non-meritocratic considerations. Because school meritocracy has been linked to inequality legitimation and broader distributive attitudes (Batruch et al., 2022; Castillo et al., 2024; Darnon et al., 2018; Wiederkehr et al., 2015), an important implication is that "meritocracy effects" are likely dimension-specific: perceptions are not equivalent to preferences, and meritocratic preferences need not have the same implications as non-meritocratic ones.

The second hypothesis stated that the measurement model would be stable over time, and the results support this expectation. Achieving strict longitudinal invariance indicates that the scale functions equivalently across waves, allowing within-student differences over time to be interpreted as substantive rather than as artifacts of item interpretation or response use (Y. Liu et

al., 2017; Putnick & Bornstein, 2016; Van De Schoot et al., 2015). This strengthens longitudinal inference and provides a firmer basis for studying developmental patterns in meritocratic beliefs, something rarely established in prior research (e.g., Chauvin et al., 2026; C. Liu & Wang, 2025; Tang et al., 2025).

The third hypothesis proposed invariance across educational levels, but the results did not support this expectation. The lack of cross-level invariance appears to be concentrated in perceived meritocracy, suggesting that students at different school stages interpret whether effort is rewarded differently (Putnick & Bornstein, 2016; Van De Schoot et al., 2012). Substantively, primary students perceive greater rewards for effort than secondary students do. A plausible explanation is that cumulative exposure to grading, selection, tracking, and broader social inequalities produces a "reality shock" in which effort appears less sufficient for success (Tang et al., 2025). This means that differences across school stages may reflect not only different levels of meritocratic belief, but also different meanings attached to meritocratic reward.

6 Conclusion

Meritocratic beliefs remain central to how inequality is interpreted, and schools are key settings in which they are socially produced and reinforced (Mijs, 2016b, 2021; Resh & Sabbagh, 2014; Sandel, 2020; Traini et al., 2025). Yet empirical research has often conflated perceptions with preferences and treated merit and non-merit as opposite poles, obscuring dual configurations (Reynolds & Xian, 2014; Wiesner & Sachweh, 2026). Extending Castillo et al.'s (2023) framework to adolescents, this study shows that students distinguish between perceived and preferred meritocracy and non-meritocracy, and that endorsement of merit can coexist with recognition of non-meritocratic advantage.

Methodologically, these findings support modeling school meritocracy as distinct yet related dimensions rather than collapsing them into a single score. The scale is strictly invariant over time for individual students, providing a robust basis for longitudinal analysis, but it does not achieve cross-level equivalence, indicating that comparisons across educational stages should be made with caution. More broadly, what is often interpreted as a change in belief in meritocracy may partly reflect increasing recognition of non-meritocratic constraints rather than declining endorsement of merit, consistent with recent accounts of dual consciousness (C. Liu & Wang, 2025; Tang et al., 2025; Zhu, 2025).

Despite its contributions, this study has limitations that point to clear avenues for future research. The scale is deliberately minimalist, with two items per factor, which constrains content coverage and reliability. The design includes only two waves and two school levels, limiting the analysis of longer developmental trajectories. In addition, cross-level non-invariance suggests the need to refine perceived-meritocracy items and test differential item functioning. Future research should expand the item pool, include additional waves and age groups, more fully distinguish types of non-meritocratic beliefs, and examine how these dimensions relate to inequality legitimation across school contexts. Overall, the study provides a stronger foundation for analyzing how school-based meritocratic socialization shapes interpretations of fairness and inequality.

7 References

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